

A Stability Partial Result for the RMF Question in Banach-Valued Kato Theory

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Source Question

The source paper is Hytonen–McIntosh–Portal, *Kato’s square root problem in Banach spaces*, arXiv:math/0703012 [1]. In Definition 2.12 they introduce the Rademacher maximal function property (RMF). Immediately after the definition they write that it would be particularly useful to know whether every UMD space has RMF, because this would let their main theorem be stated for all UMD spaces.

The relevant source passage is on page 8 of the arXiv PDF:

spaces, we need to introduce a new class of spaces, the denning property of which is the boundedness of the following *Rademacher maximal function*:

$$M_R u(x) := \sup \left\{ \mathbb{E} \left\| \sum_{k \in \mathbf{Z}} \varepsilon_k \lambda_k A_{2^k} u(x) \right\|_X : \right. \\ \left. \lambda = (\lambda_k)_{k \in \mathbf{Z}} \text{ finitely non-zero with } \|\lambda\|_{\ell^2(\mathbf{Z})} \leq 1 \right\}.$$

Note that, under the identification $X \simeq \mathcal{L}(\mathbf{C}, X)$, this is the R -bound of the set

$$\{A_{2^k} u(x) : k \in \mathbf{Z}\} = \{\langle u \rangle_Q : Q \ni x\}.$$

In particular, if X is a Hilbert space, we recover the usual dyadic maximal function.

Definition 2.12. We say that the Banach space X has the RMF property, if M_R is bounded from $L^2(\mathbf{R}^n; X)$ to $L^2(\mathbf{R}^n)$.

We do not yet completely understand how this new class of spaces relates to the other Banach space notions discussed above, which forces us to adopt this property as an additional assumption. It would be particularly useful to know if every UMD space has RMF, since this would allow us to state our main theorem in the generality of all UMD spaces, but the question remains open. However, in Section 7 we show that the RMF property does hold in most of the concrete situations of interest. The classes of Banach spaces appearing in the statement are also defined in Section 7.

Result Type and Scope

This packet gives a partial result, not a full answer to the UMD question. It proves that RMF is preserved by three standard Banach-space operations: closed subspaces, quotients, and complex interpolation. Therefore the positive part of the source question extends from the concrete classes listed in [1] to the subquotient and interpolation hull generated by those classes.

The full question remains open here: the packet does not prove that every UMD space lies in this hull, and it does not construct a UMD space without RMF.

Definitions

For a finite set J of dyadic levels and a Banach-space-valued function f , write $E_j f$ for the dyadic conditional expectation at level j . The finite Rademacher maximal function is

$$M_{R,J}^X f(\omega) = \sup_{\|\lambda\|_{\ell^2(J)} \leq 1} \mathbb{E}_\varepsilon \left\| \sum_{j \in J} \varepsilon_j \lambda_j E_j f(\omega) \right\|_X.$$

The space X has RMF if, for some (equivalently all) $1 < p < \infty$, the quantities $M_{R,J}^X$ are bounded on $L^p(\Omega; X)$, uniformly over finite J , for the dyadic filtration. This is equivalent to the usual formulation used in [1]; Kemppainen's work shows that the property is independent of the particular underlying filtration in the standard sense [2].

Proof Intuition

The Rademacher maximal function is nonlinear only because one takes a pointwise operator norm over finite Rademacher sums. At each finite truncation J , it is the norm of a linear operator

$$f \mapsto [\lambda \mapsto \sum_{j \in J} \varepsilon_j \lambda_j E_j f(\omega)].$$

This linearization makes stability almost formal. Subspaces are immediate because averages remain in the subspace. Quotients work because conditional expectation commutes with quotient maps and Bochner L^p quotients allow almost norm-preserving liftings. Interpolation works by applying complex interpolation to the finite linearized operators, using the standard interpolation identity for finite Rademacher spaces. The constants are independent of J , so the finite estimates pass to the full maximal function.

Theorem

Theorem 1 (RMF stability). *Let $1 < p < \infty$.*

1. *If X has RMF and $Y \subset X$ is a closed subspace, then Y has RMF.*
2. *If X has RMF and $Q : X \rightarrow Z$ is a quotient map onto a Banach space Z , then Z has RMF.*
3. *If (X_0, X_1) is a compatible complex interpolation couple and both X_0 and X_1 have RMF, then $X_\theta = [X_0, X_1]_\theta$ has RMF for every $0 < \theta < 1$.*

Proof. We use the finite quantities $M_{R,J}$; the full RMF estimate follows by monotone convergence over increasing finite sets J .

For subspaces, let $f \in L^p(\Omega; Y)$. Since the dyadic conditional expectations of f are still Y -valued and the norm on Y is inherited from X , one has

$$M_{R,J}^Y f(\omega) = M_{R,J}^X f(\omega).$$

The L^p estimate for Y follows from the one for X .

For quotients, normalize Q to be the metric quotient map $X \rightarrow X/Y = Z$; the general quotient case is the same after the usual equivalent normalization. The Bochner quotient identity

$$L^p(\Omega; Z) \cong L^p(\Omega; X)/L^p(\Omega; Y)$$

implies that for every $f \in L^p(\Omega; Z)$ and every $\delta > 0$ there is $g \in L^p(\Omega; X)$ with $Qg = f$ and

$$\|g\|_{L^p(\Omega; X)} \leq (1 + \delta)\|f\|_{L^p(\Omega; Z)}.$$

Conditional expectation commutes with the bounded linear map Q , so

$$E_j^Z f = QE_j^X g.$$

Therefore, for every ω ,

$$M_{R,J}^Z f(\omega) \leq M_{R,J}^X g(\omega).$$

Taking L^p -norms, using RMF for X , and then letting $\delta \downarrow 0$, proves RMF for Z .

It remains to prove interpolation stability. For a finite set J , define the linearized operator

$$T_J^X f(\omega) : \ell^2(J) \rightarrow \text{Rad}_J(X), \quad T_J^X f(\omega)\lambda = \sum_{j \in J} \varepsilon_j \lambda_j E_j f(\omega).$$

Then

$$\|T_J^X f(\omega)\|_{\mathcal{L}(\ell^2(J), \text{Rad}_J(X))} = M_{R,J}^X f(\omega).$$

If X_i has RMF constant C_i on L^p , then

$$T_J^{X_i} : L^p(\Omega; X_i) \longrightarrow L^p(\Omega; \mathcal{L}(\ell^2(J), \text{Rad}_J(X_i)))$$

has norm at most C_i , uniformly in J , for $i = 0, 1$.

Complex interpolation gives

$$[L^p(\Omega; X_0), L^p(\Omega; X_1)]_\theta = L^p(\Omega; [X_0, X_1]_\theta).$$

We also use the standard finite Rademacher interpolation identity

$$\text{Rad}_J([X_0, X_1]_\theta) = [\text{Rad}_J(X_0), \text{Rad}_J(X_1)]_\theta$$

with constants independent of J . Finally, evaluation on the common domain $\ell^2(J)$ gives the contractive embedding

$$[\mathcal{L}(\ell^2(J), \text{Rad}_J(X_0)), \mathcal{L}(\ell^2(J), \text{Rad}_J(X_1))]_\theta \hookrightarrow \mathcal{L}(\ell^2(J), \text{Rad}_J([X_0, X_1]_\theta)).$$

Interpolating the endpoint bounds for T_J , and then applying this embedding, yields

$$\|M_{R,J}^{X_\theta} f\|_{L^p(\Omega)} \leq C_0^{1-\theta} C_1^\theta \|f\|_{L^p(\Omega; X_\theta)}$$

up to the harmless universal constants in the Rademacher interpolation identity. The bound is independent of J , so X_θ has RMF. $\square$

Consequence for the Source Problem

Corollary 1. *Let $\mathcal{C}$ be the smallest class of Banach spaces containing the RMF examples listed in [1] and closed under closed subspaces, quotients, and complex interpolation. Every space in $\mathcal{C}$ has RMF. Hence the Banach-valued Kato theorem of [1] applies to every UMD space X such that both X and X^* belong to $\mathcal{C}$.*

Proof. Iterate Theorem 1. The final statement is exactly the RMF hypothesis needed in the main theorem and corollaries of [1]. $\square$

Verification Notes

No computational search is needed for the proof. The main proof dependencies are standard: Bochner L^p quotients, interpolation of Bochner L^p spaces, interpolation of finite Rademacher spaces, and the elementary interpolation embedding for operator spaces with a common domain. The reviewer should check these standard functional-analytic ingredients, especially that the constants in the finite Rademacher interpolation identity are independent of the finite set J .

Novelty and Literature Check

Before promotion, the run indexes were searched for 0703012, Kato, square root, RMF, Rademacher maximal, and UMD. No prior packet or attempt for arXiv:math/0703012 was found. Local source arXiv:1002.2876 records the RMF-vs-UMD problem as still open in 2010 and proves Carleson-embedding equivalences for RMF [3]. Web searches on June 14, 2026 used the phrases “Rademacher maximal function RMF UMD counterexample”, “every UMD space RMF”, and “RMF property interpolation Banach spaces”. They found later multilinear/operator-valued work using or avoiding RMF assumptions, but no exact full solution of the UMD question and no duplicate of this stability packet.

Limitations

This is only a closure theorem. It does not prove that arbitrary UMD spaces can be constructed from the known RMF examples using subspaces, quotients, and complex interpolation. It also does not rule out a UMD counterexample outside this closure class.

References

References

- [1] T. Hytönen, A. McIntosh, and P. Portal, *Kato’s square root problem in Banach spaces*, arXiv:math/0703012, 2007.
- [2] M. Kempainen, *On the Rademacher maximal function*, arXiv:0912.3358, 2009.
- [3] T. Hytönen and M. Kempainen, *On the vector-valued Carleson embedding theorem and the Rademacher maximal function*, arXiv:1002.2876, 2010.